

APPENDIX B

Probability and statistics

B.1. Space story

I met Professor Van der Velde at MIT to talk about estimation. I was thinking of taking his estimation course. He told me that he always gave out a list of 100 probability problems for students to do as a review for the course as it was assumed that graduate students had some background in the material. He said that each year, without fail, the students could hardly do any of the problems despite having taken courses in probability and statistics.

B.2. Introduction

Probability and statistics are a critical aspect of attitude control and determination systems. They enter into the discipline in many ways. For example,

1. Sensor measurements have noise.
2. Certain disturbances on a spacecraft are best characterized as noise.
3. Alignments of sensors and actuators on the spacecraft are only known with a degree of uncertainty. This must be considered in pointing budgets.
4. Actuators, sensors, and supporting hardware have a probability of failure and an expected lifetime.
5. When landing on a planet, moon, or asteroid, the surface is not known perfectly. There is some probability that the landing spot will have a boulder or be too soft to support the lander.
6. For a debris-removal satellite, the distribution of debris is not known perfectly and the angular rates of the debris may not be known.
7. Link budgets, which provide requirements, are impacted by the signal-to-noise ratio of the telecommunications systems. Noise is a random process.

This appendix discusses concepts in probability and statistics that are needed for aircraft control-system design. In particular, it discusses stochastic processes that are used for modeling gusts and other phenomena, and the characterization of noise models for sensors. Stochastic processes appear in several areas of aircraft control:

1. External disturbances: wind and wind gusts;
2. Noise: sensor noise;
3. Tolerances: mechanical tolerances on sensor alignments, actuator alignments, actuator and sensor scale factors, etc.

B.3. Axiomatic probability

The basic notion of probability is

$$P(A) = \frac{\text{Number Of Outcomes With A}}{\text{Total Outcomes}} \quad (\text{B.1})$$

For a coin flip there are two outcomes, heads or tails and one outcome favoring heads therefore the probability of heads is 0.5. This definition assumes that all outcomes are equally likely. A more general method of approaching probability is an axiomatic probability that deals with sample spaces. A sample space is the set of all possible outcomes. Each outcome is called an element or point in the sample space. The number of points in the sample space may be finite, countably infinite or infinite. For example, in the experiment of drawing from a deck of cards, there are 52 possible outcomes. Thus the sample space consists of a list of all of the cards in the deck. The axioms of probability are that the probability of any outcome is greater than or equal to zero. The probability that at least one outcome will occur is 1 and if the events are mutually exclusive the probability of any set of the events occurring is equal to the sum of the probabilities of each event occurring. In equation form, the probabilities are

$$P(A) \geq 0 \quad (\text{B.2})$$

$$P(S) = 1 \quad (\text{B.3})$$

$$P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3) \quad (\text{B.4})$$

where $\cup$ is the union symbol and means combining the sets. Sometimes you want to know the probability of one outcome given that another has already occurred. This is the conditional probability and is given as

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (\text{B.5})$$

where $\cap$ is the intersection of the two sets. This leads to Bayes rule, which is

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \quad (\text{B.6})$$

Events A and B are independent if

$$P(A \cap B) = P(A)P(B) \quad (\text{B.7})$$

If this is the case the conditional probability is

$$P(A|B) = \frac{P(B)P(A)}{P(B)} = P(A) \quad (\text{B.8})$$

B.4. Binomial theorem

The binomial theorem

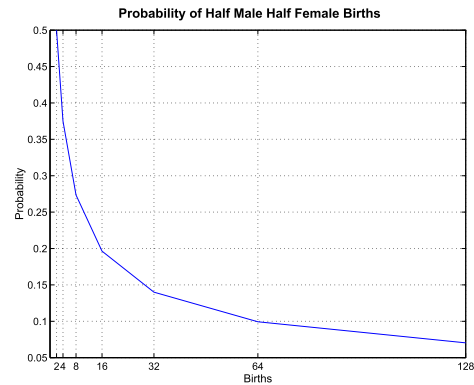
$$p(k|n) = \frac{n!}{(n-k)!k!} p^k (1-p)^{n-k} \quad (\text{B.9})$$

gives the probability of k occurrences in n samples if the probability of one occurrence is p . For example suppose two dogs are born. What is the probability that one is female and one male given that the probability of a male is 0.5? This can be done by simple counting. The four possibilities are (where 0 is male and 1 is female):

00
01
10
11

There are four possibilities of which two have one male and one female. Thus the probability is 0.5. However, for larger numbers of samples the probability of there being half males and half males decreases because the number of cases in which there are different numbers of males and females grows in relation to the number of cases where there are half males and half females. See Example B.1.

```
1 births = [2 4 8 16 32 64 128];
2
3 pC = zeros(1,7);
4 for j = 1:length(births)
5     pC(j) = BinomialTheorem( births(j), births(j)
6         /2, 0.5 );
7 end
8 Plot2D( births , pC, 'Births', 'Probability' ,...
9     'Probability of Half Male Half Female Births')
10 set( gca, 'xlim', [0 births(end)], 'xtick',
    births )
```



Example B.1: Dog-birth problem.

The binomial theorem can be used for hypothesis testing. Suppose that a gun fires 100 shots and hits 50 times. What is the probability that the gun is 50% accurate? We write

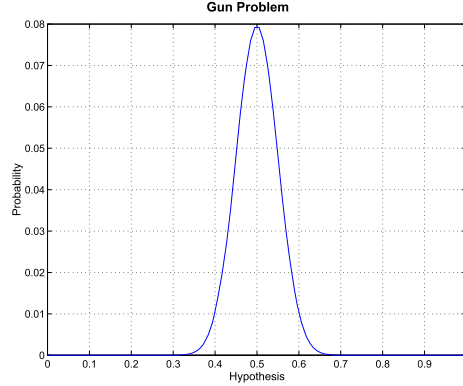
$$p(k|n) = \frac{100!}{50!50!} 0.5^k (0.5)^{n-k} \quad (\text{B.10})$$

which is about 8%. $!$ is the factorial symbol $3! = 3 \times 2 \times 1$. This seems low, but Example B.2 shows that this is the most likely hypothesis. Thus the probability of the hypothesis being true must be compared to the probability of any other hypothesis being true.

```

1 p = linspace(0,1);
2
3 pC = zeros(1,length(p));
4 for j = 1:length(p)
5     pC(j) = BinomialTheorem( 100, 50, p(j) );
6 end
7
8 Plot2D( p, pC, 'Hypothesis', 'Probability', 'Gun_
    Problem' )

```



Example B.2: Hypothesis testing.

B.5. Probability distributions

The cumulative probability-distribution function is written as

$$F(x) = P[X \leq x] \quad (\text{B.11})$$

$$F(\infty) = 0 \quad (\text{B.12})$$

This says that the probability that X is between $\pm\infty$ is 1. $F(x)$ is the probability that the value of the random variable X is less than x . The probability-density function (pdf) is the derivative of the cumulative probability-distribution function or

$$f(x) = \frac{d}{dx} F(x) \quad (\text{B.13})$$

and

$$F(x) = \int_{-\infty}^x f(x) dx \quad (\text{B.14})$$

If the integral equals 1, then the probability that X is between two values a and b is

$$P[a < X \leq b] = F(b) - F(a) \quad (\text{B.15})$$

$$= \int_{-\infty}^b f(x) dx - \int_{-\infty}^a f(x) dx = \int_a^b f(x) dx \quad (\text{B.16})$$

The mean, or expected value E , is

$$E(X) = \mu = \int_{-\infty}^{\infty} x f(x) dx \quad (\text{B.17})$$

The root mean-squared value is

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx \quad (\text{B.18})$$

The variance is

$$\text{Var}(X) = \sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx \quad (\text{B.19})$$

so that

$$\sigma = E(X^2) - \mu^2 \quad (\text{B.20})$$

The variance of a random variable multiplied by a scalar can be easily found and is

$$\text{Var}(aX) = a^2 \text{Var}(X) \quad (\text{B.21})$$

If random variables are independent, that is the value of one does not depend on the value of another (such as the flipping of two points), the mean value of the product is

$$E(X_1 X_2) = E(X_1) E(X_2) \quad (\text{B.22})$$

and the variance of the sum is

$$\text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2) \quad (\text{B.23})$$

An estimate of the mean value of X can be found by taking N samples

$$\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i \quad (\text{B.24})$$

The sample variance is

$$S^2 = \frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X})^2 \quad (\text{B.25})$$

Note that if instead of $N-1$ in the denominator we had N we would be computing the population variance. The sample standard deviation is

$$S = \sqrt{S^2} \quad (\text{B.26})$$

Note that each value in the sum is in itself a random variable since each measured value is subject to chance. Thus

$$E(\bar{X}) = E\left(\frac{1}{N} \sum_{i=1}^N X_i\right) = \frac{1}{N} \sum_{i=1}^N E(X_i) = \frac{N\mu}{N} \quad (\text{B.27})$$

Thus the sample mean is an unbiased estimate of the mean value m . Likewise $E(S^2) = \sigma^2$. In the limit, as you sum a large number of independent random variables you get a Gaussian or normal distribution. The probability density function for the Gaussian distribution is

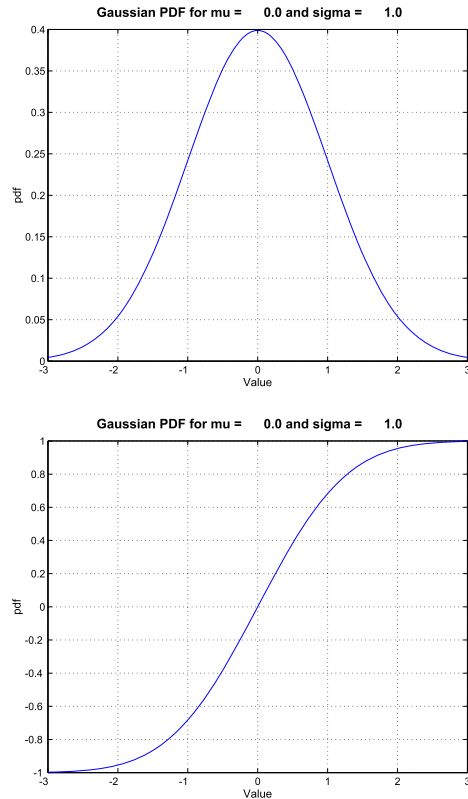
$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (\text{B.28})$$

With a Gaussian distribution 68% of the values will fall within $\pm\sigma$, 95% of the values will fall within $\pm 2\sigma$, and 99.7% of the values will fall within $\pm 3\sigma$. The Gaussian pdf and cpdf are shown in Example B.3. Both figures are over the same 3σ range. By the way, 6σ is 99.999998.

```

1 x = linspace(-3,3);
2 mu = 0;
3 sigma = 1;
4 pDF = GaussianPDF( x, mu, sigma );
5 cPDF = GaussianCPDF( x, mu, sigma );
6 t = sprintf('Gaussian PDF for mu = %8.1f and sigma = %8.1f', mu, sigma);
7 Plot2D( x, pDF, 'Value', 'pdf', t );
8 Plot2D( x, cPDF, 'Value', 'pdf', t );

```



Example B.3: Gaussian pdf and cpdf.

B.6. Evaluating measurements

Two measures for evaluating measurements are the confidence interval and sign test. The confidence interval is based on the Gaussian pdf and tells you the probability that a value will lie with $x\%$ of the nominal. A way of comparing two identical instruments is the sign test. Given two sequences of measurements done under identical circumstances we subtract one set from the other and count positive signs. If the errors are truly random then half the signs will be positive and the other half negative.

B.7. Combining errors

If a quantity can be expressed in the form

$$y = f(x_1, x_2) \quad (\text{B.29})$$

then the contribution of errors in x_1 and x_2 to y can be found using the following formulas. Let

$$a = \left[\frac{\partial f}{\partial x_1} \Big|_x \frac{\partial f}{\partial x_2} \Big|_x \right]^T \quad (\text{B.30})$$

where x is the current or nominal value of the state vector. Then, the error estimate is

$$z = \frac{xa}{y} \quad (\text{B.31})$$

$$e_{abs} = \sum_x |z_k| e_k \quad (\text{B.32})$$

$$e_{SRSS} = \sqrt{\sum_k (z_k e_k)^2} \quad (\text{B.33})$$

where e is the fractional error for each element of x . xa is

$$xa = \begin{bmatrix} a_1 x_1 & a_2 x_2 & \dots & a_N x_N \end{bmatrix} \quad (\text{B.34})$$

The first error is more conservative than the second.

B.8. Multivariate normal distributions

The multivariate normal distribution is important in many areas of control. It is defined as

$$f(x) = \frac{1}{(s\pi)^{n/2} \sqrt{|C|}} e^{-\frac{1}{2}(x-\mu)^T C^{-1}(x-\mu)} \quad (\text{B.35})$$

where C is the covariance matrix and is

$$C = \begin{bmatrix} E[(X_1 - \mu_1)^2] & E[(X_1 - \mu_1)(X_2 - \mu_2)] \\ E[(X_1 - \mu_1)(X_2 - \mu_2)] & E[(X_2 - \mu_2)^2] \end{bmatrix} \quad (\text{B.36})$$

$$C = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix} \quad (\text{B.37})$$

for a two-dimensional system. σ is the variance and ρ the correlation coefficient. The determinant in this case is

$$|C| = (1 - \rho^2)\sigma_1^2\sigma_2^2 \quad (\text{B.38})$$

which is zero if $\rho = 1$ because the two variables are completely correlated, i.e., indistinguishable and not independent.

B.9. Random signals

A random signal is any signal that is not deterministic. The following is a deterministic signal

$$x(t) = \sin(t) \quad (\text{B.39})$$

where $x(t)$ is exactly determined as a function of t . A random signal might be one in which the amplitude is random

$$x(t) = A \sin(t) \quad (\text{B.40})$$

This would be the equivalent of having a sine-wave generator in a lab in that the amplitude of the wave was determined by a dial that is set to the value that the last engineer used. When you turn on the generator you would get a sine wave with a constant amplitude but the value of that amplitude would be random. There are two important definitions. Assume that you run an experiment in which you measure a signal. You then run the same experiment many times. This produces an ensemble of time signals. If averaging over a single time signal is the same as averaging over the ensemble of time signals then the process is ergodic. If the probability density functions do not change as a function of time, the process is time stationary. The autocorrelation function for a random process $X(t)$ is

$$R_X(T_1, t_2) = E[X(t_1)X(t_2)] \quad (\text{B.41})$$

If the process is stationary

$$R_X(\tau) = E[X(t)X(t + \tau)] \quad (\text{B.42})$$

The crosscorrelation is written as

$$R_{XY}(\tau) = E[X(t)Y(t + \tau)] \quad (\text{B.43})$$

for a stationary process. The power spectral density is the Fourier transform of the crosscorrelation and is

$$S_X(j\omega) = \int_{-\infty}^{\infty} R_X(\tau) e^{-j\omega\tau} d\tau \quad (\text{B.44})$$

The cross-spectral density is defined in a similar fashion. White noise is defined as a stationary random process having a constant spectral density function. Thus

$$S_{X(j\omega)} = A \quad (\text{B.45})$$

and the crosscorrelation is

$$R_X(\tau) = A\delta(\tau) \quad (\text{B.46})$$

where $\delta(\tau)$ is the unit impulse, meaning that the value at any one time is unrelated to the value at any other time.

B.10. Outliers

If we assume that the noise-corrupting data point is Gaussian, the probability that a noisy point lies within a distance of its true point decreases rapidly with the distance. For the Gaussian hypothesis to be correct, noise should not move points more than a few standard deviations. If a small percentage of points is greater than 2 or 3 standard deviations then the resulting estimate will be poor since the data is not conforming to the hypothesis on which the least-squares estimator is based. For this reason, outliers should be eliminated from the data sample.

B.11. Noise models

Random variables are needed when simulating spacecraft. They are used for sensor-noise models and for Monte Carlo simulations. Monte Carlo simulations are when you run a simulation numerous times with randomly distributed initial conditions and randomly distributed parameters. Uniform random numbers and normal random numbers are easily obtained. Some noise models require other probability distributions. A general and simple technique is the inverse cumulative function (CDF) sampling technique [1]. A continuous CDF is a one-to-one mapping of the domain of the CDF into the interval from zero to one. Knowing the inverse CDF allows us to go backwards from a random-number generator that produces a uniform distribution between 0 and 1 and

any other distribution. For example, the CDF of an exponential distribution is

$$f(t) = \theta e^{-\theta t} \quad (\text{B.47})$$

The inverse CDF is

$$F^{-1}(u) = -\frac{\log(1-u)}{\theta} \quad (\text{B.48})$$

where u is a random number between zero and 1. θ is the scaling.

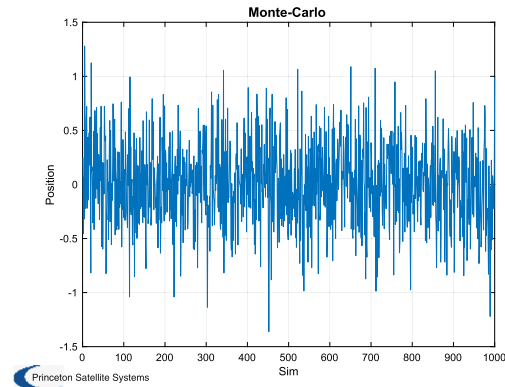
B.12. Monte Carlo methods

Monte Carlo methods, (named after the famous casino) are computational algorithms that rely on random sampling to obtain results. For example, a Monte Carlo approach to a damped second-order system simulation would be to use randomly generated step inputs to determine the performance of the system. Example B.4 shows a Monte Carlo simulation of a damped second-order system in which the input is a Gaussian random number and the damping ratio is a uniform random number between 0 and 1. This might be used for a slosh study where the damping is not well known.

```

1 %% Monte-Carlo simulation
2
3 n = 1000;
4 xP = zeros(1,n);
5 dT = 0.1;
6
7 for j = 1:n
8     x = [0;0];
9     a = 0.01*randn;
10    zeta = rand;
11    for k = 1:100
12        x = RK4(@RHS,x,dT,0,a,zeta);
13    end
14    xP(1,j) = x(1);
15 end
16
17 Plot2D(1:n,xP,'Sim','Position','Monte-Carlo');
18
19
20 function xDot = RHS(x,~,a,zeta)
21
22 omega = 0.1;
23
24 xDot = [x(2); -zeta*omega*x(2) - omega^2*x(1) + a
25         ];
26 end

```



Example B.4: Monte Carlo simulation with a Gaussian random input and a uniform random-number damping ratio.

References

- [1] R. Wicklin, *Simulating Data with SAS*, SAS Institute, April 2013.